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# Nicholas M. Rapidis

Curriculum Vitae


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## Academic Positions

2026- **Yale University**
Incoming Yale Quantum Institute Fellow
Incoming postdoctoral researcher in Prof. Konrad Lehnert's group developing superconducting sensors for readout of electromagnetic systems, such as dark matter axion experiments.

## Education

2019-present **Stanford University**
Ph.D. Candidate in Physics. Expected graduation: Spring 2026
Advisor: *Prof. Kent Irwin*
M.S. in Physics completed in July 2022

2015-2019 **University of California, Berkeley**
B.A. in Physics. Graduated with High Distinction in General Scholarship and Departmental Honors
Thesis Title: *Resonant Axion-Photon Scattering and Galactic Searches for Axions*
Advisor: *Prof. Karl van Bibber*

## Research Experience

2020-present **Graduate Research Assistant**, Stanford University
Advisor: *Prof. Kent Irwin*
Member of the Dark Matter Radio (DMRadio) collaboration. Presented DMRadio-Core. Presented design concept and first models of DMRadio-GUT. Lead on the design, modeling, and optimization of science reach of DMRadio-m$^3$. Lead on DMRadio-50L SQUID readout chain, superconducting sheath design, and contributions to other cryogenic hardware components. Former lead on SQUID testing for BICEP array.

2019-2020 **Graduate Research Assistant**, Stanford Institute for Theoretical Physics
Advisor: *Prof. Savas Dimopoulos*
Studied phenomenology of dense dark matter axion clumps (oscillons) collisions with neutron stars. Set limits on oscillon dark matter abundance.

2016-2019 **Undergraduate Research Assistant**, UC Berkeley

Advisor: *Prof. Karl van Bibber*

Member of the *Haloscope at Yale Sensitive to Axion Cold Dark Matter (HAYSTAC)* collaboration. Co-lead on refurbishment and optimization for cavity used HAYSTAC Phase II. Introduced extensive use of finite element simulation techniques for axion cavity characterization.

## Honors & Awards

2022 *Young Scientist Award* at *Identification of Dark Matter 2022*: One of best three talks (out of 90) given by graduate students and postdocs at the conference.

2019 *Phi Beta Kappa*

2018-2019 UC Berkeley *Haas Scholar*

## Professional Activities

2023- Journal Referee: *Physical Review Letters*, *Physical Review D*, *Journal of Low Temperature of Physics*

## Publications

[inSPIRE Profile] [Google Scholar Profile]

### Principal Author

[18] **DMRadio-Core: A new approach to GUT-scale axion searches**
V. Ankel *et al.* [arXiv:2604.16602][inSPIRE]

[17] **Electromagnetic modeling and science reach of DMRadio-m$^3$**
A. AlShirawi *et al.* *Phys. Rev. D* **112**, 052001 (2025) [arXiv:2302.14084][inSPIRE]

[16] **Characterization of the HAYSTAC axion dark matter search cavity using microwave measurement and simulation techniques**
N.M. Rapidis *et al.* *Review of Scientific Instruments* **90**, 024706 (2019) [arXiv:1809.02246][inSPIRE].

### Joint Principal Author

[15] **Resonant Conversion of Dark Matter Oscillons in Pulsar Magnetospheres**
A. Prabhu and N.M. Rapidis, *JCAP* **10**, (2020) 054 [arXiv:2005.03700][inSPIRE].

### Contributing Author

[14] **Search for Dark Photons between 16.96 – 19.52 $\mu$eV with the HAYSTAC experiment**
X. Bai *et al.* *Phys. Rev. D* **113**, 012010 (2026) [arXiv:2510.15848] [inSPIRE]

[13] **Noise limits for dc SQUID readout of high-$Q$ resonators below 300 MHz**
V. Ankel *et al.* *J. Appl. Phys.* **138**, 094505 (2025) [arXiv:2504.20398] [inSPIRE]

[12] **Dark Matter Axion Search with HAYSTAC Phase II**
X. Bai *et al.* *Phys. Rev. Lett.* **134**, 151006 (2025) [arXiv:2409.08998] [INSPIRE]

[11] **Measurements of DC SQUID Damping Effects on Superconducting Resonant Circuits**
E. C. van Assendelft *et al.* *IEEE Transactions on Applied Superconductivity* **33**, 5, (2023) [INSPIRE]

[10] **New Results from HAYSTAC's Phase II Operation with a Squeezed State Receiver**
M.J. Jewell *et al.* *Phys. Rev. D* **107**, 072007, (2023) [arXiv:2301.09721][INSPIRE]

[9] **Quantum metrology of low frequency electromagnetic modes with frequency upconverters**
S.E. Kuenstner *et al.* Phys. Rev. Research **7**, 013281 (2025)[arXiv:2210.05576][INSPIRE]

[8] **DMRadio-m$^3$: A Search for the QCD Axion Below 1 $\mu$eV**
L. Brouwer *et al.* *Phys. Rev. D* **106**, 103008, (2022) [arXiv:2204.13781][INSPIRE]

[7] **Introducing DMRadio-GUT, a search for GUT-scale QCD axions**
L. Brouwer *et al.* *Phys. Rev. D* **106**, 112003, (2022) [arXiv:2203.11246][INSPIRE]

[6] **A Model-Independent Radio Telescope Dark Matter Search**
A. Keller, *et al.* *Astrophys. J.* **927** (2022) 1, 71. [arXiv:2112.03439][INSPIRE]

[5] **A quantum-enhanced search for dark matter axions**
K.M. Backes *et al.* *Nature* **590**, 238-242 (2021) [arXiv:2008.01853][INSPIRE]

[4] **An improved analysis framework for axion dark matter searches**
D.A. Palken *et al.* *Phys. Rev. D* **101**, 123011, (2020) [arXiv:2003.08510][INSPIRE].

[3] **Results from Phase 1 of the HAYSTAC microwave cavity axion experiment**
L. Zhong *et al.* *Phys. Rev. D* **97**, 092001, (2018) [arXiv:1803.03690][INSPIRE].

[2] **Design and Operational Experience of a Microwave Cavity Axion Detector for the 20-100 $\mu$eV Range**
S. Al Kenany *et al.* *Nuclear Instruments and Methods in Physics Research A* **854** (2017) 11–24. [arXiv:1611.07123] [INSPIRE].

[1] **First Results from a Microwave Cavity Axion Search at 24 $\mu$eV**
B.M. Brubaker *et al.* *Phys. Rev. Lett.* **118**, 061302 (2017) [arXiv:1610.02580][INSPIRE].

# Talks

## Invited Talks

**GUT scale axion searches with DMRadio**
SLAC National Accelerator Laboratory, Elementary Particle Physics Theory Seminar, December 10, 2025

**Searching for sub-$\mu$eV axions below the standard quantum limit**
Yale University, Yale Quantum Institute Seminar, December 1, 2025

**Searching for low mass dark matter axions with DMRadio**
Boston University, Physics Seminar, July 24, 2024

**Status of the DMRadio Program**
*YOUNGST@RS - Shoot for the Stars, Aim for the Axions*, October 4-7, 2022, Virtual.

## Selected Contributed Talks

**DMRadio-GUT: quantum advantage at the GUT scale**
*20th Patras Workshop on Axions, WIMPs, and WISPs*, September 22-26, 2025, Tenerife, Spain

**Status of DMRadio-50L and m$^3$**
*Identification of Dark Matter*, July 18-22, 2022, Vienna, Austria

**Completion of Phase I and Preparation for Phase II of the HAYSTAC Experiment**
*14th Patras Workshop on Axions, WIMPs, and WISPs*, June 18-22, 2018, DESY, Hamburg, Germany

## Teaching Experience

**Head Teaching Assistant**, Stanford University

Spr. 2022 Physics 25 – Modern Physics (Instructor: Kent Irwin).

2021-2022 **Mentor**, Polygence

One-on-one mentoring of high school students on research projects. Projects topics in dark matter physics and cosmology.

**Teaching Assistant**, Stanford University

Fall 2020 Physics 46 – Heat and Optics (Instructor: Giorgio Gratta).

Spr. 2020 Physics 43 – Electricity and Magnetism (Instructor: Mark Kasevich).